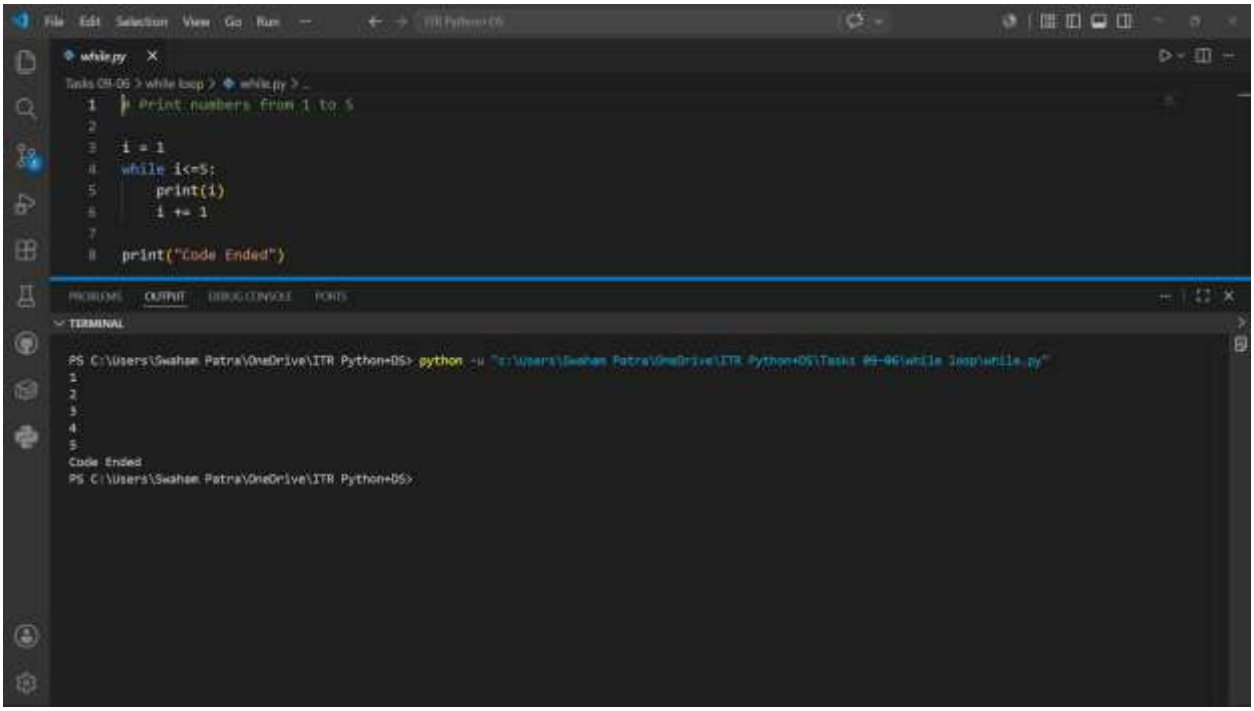
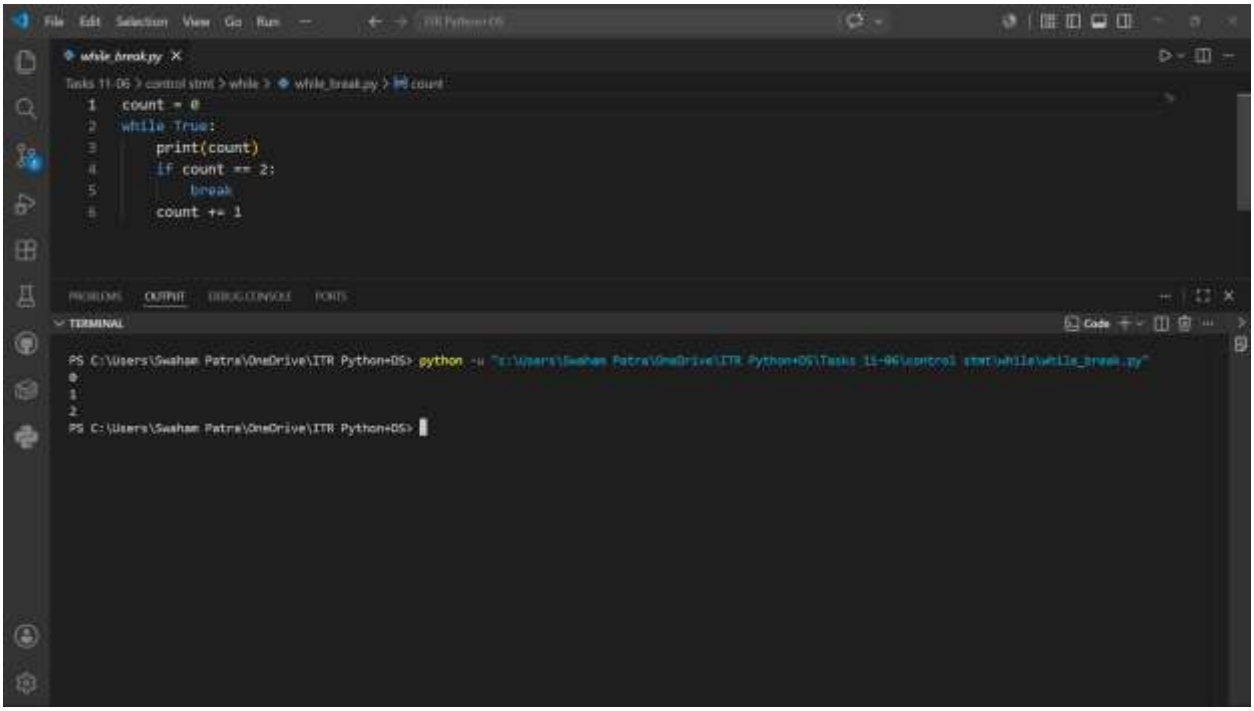
Swaham Pitambar Patra

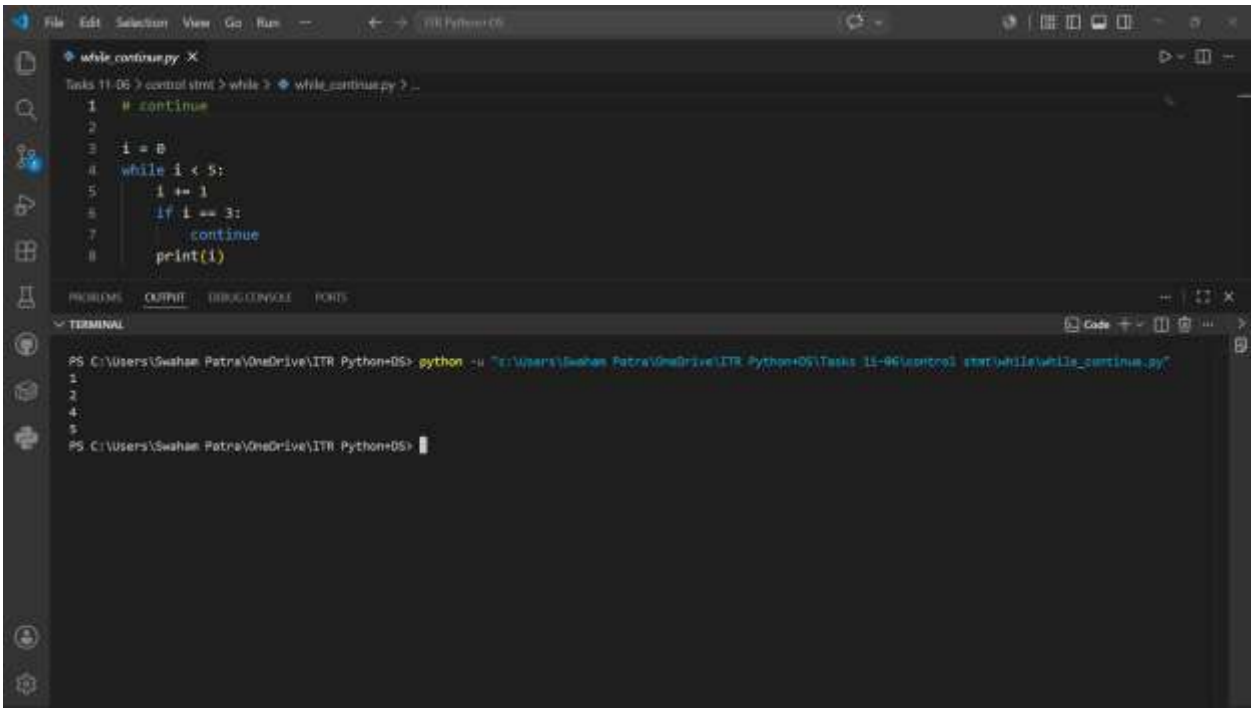
✅ Basic While Loop



✅ While with Break

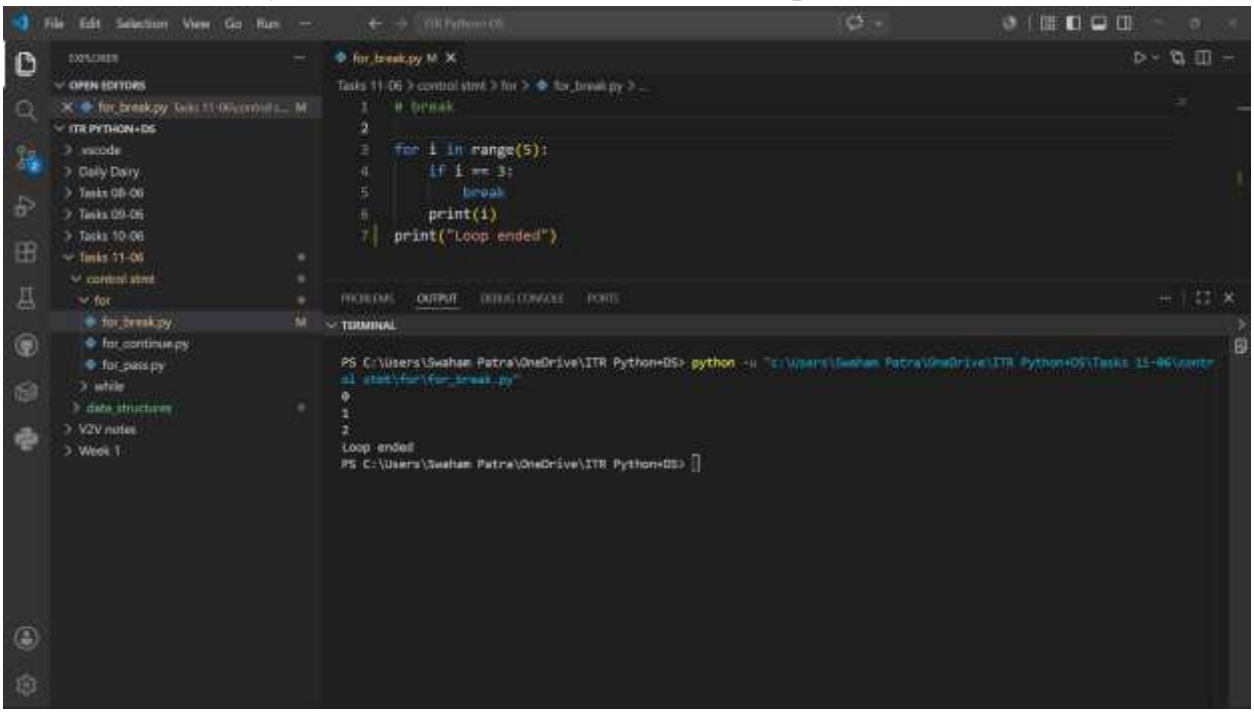


✅ While with Continue



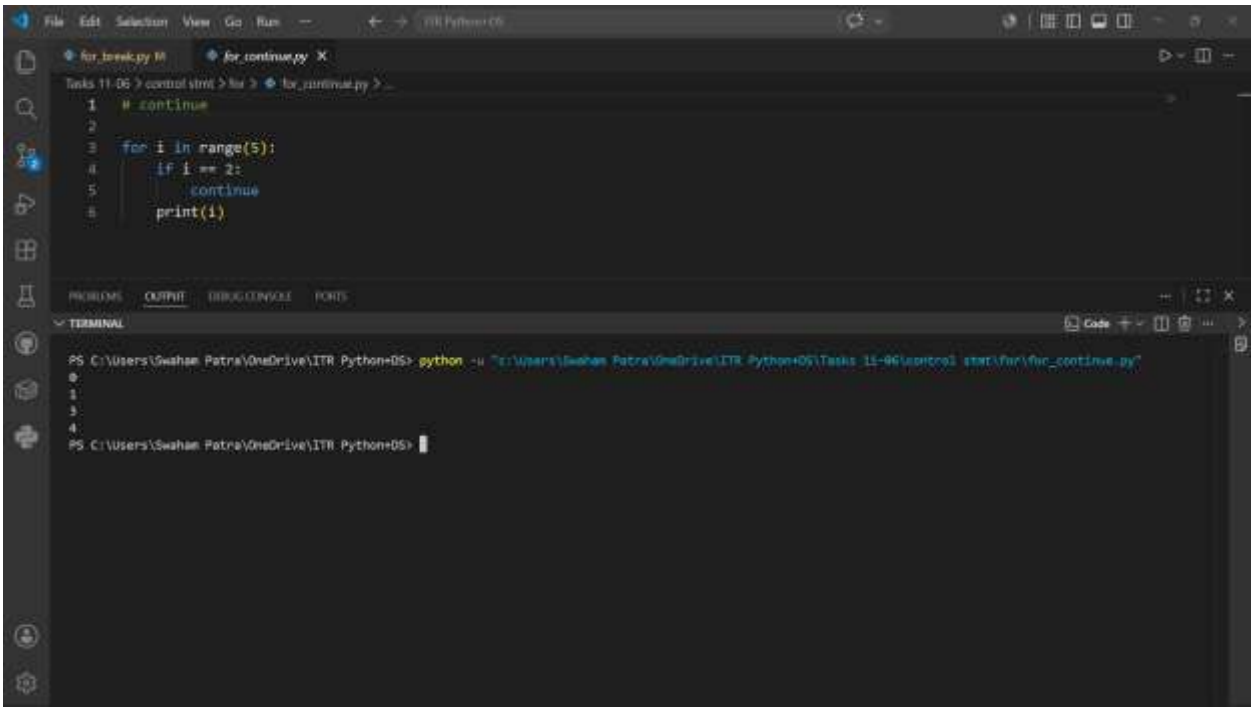
✅ Use of Break

- Used to break out of the loop after a particular condition is satisfied
  And directly execute the next line after the loop



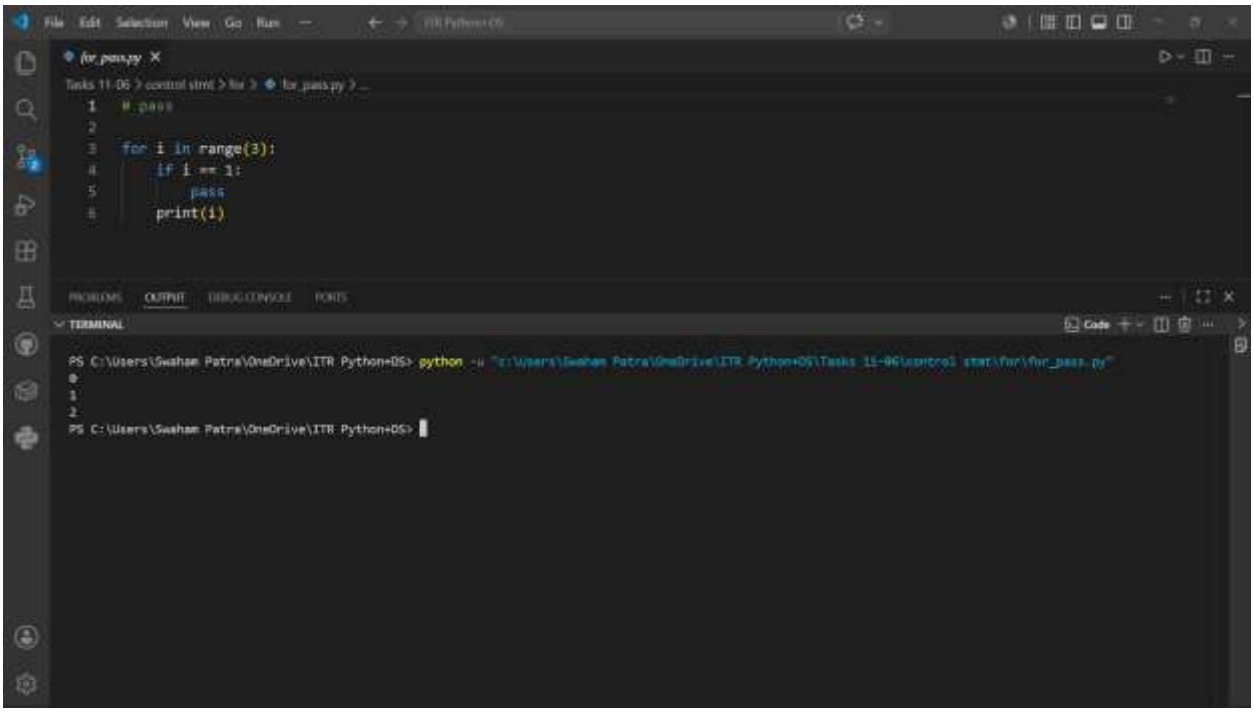
✅ Use of Continue

- Used to skip an iteration in the loop and the rest of the loop runs until the stopping point justified.



✅ Use of Pass

- Just used as a placeholder. Doesn't affect the code in any way.



✅ Use of List

- Lists are used when there is a need to make a collection of heterogeneous data in a single variable and is to be mutable.

Eg. List1 = [“person1”,……]

✅ Use of Tuple

- Tuples are used when there is a need of heterogeneous data to be in a single variable and is immutable.

Eg. Tuple1 = {10, 20,……..}

✅ Use of Sets

- Sets are used when there is a requirement of only unique heterogeneous data.

Eg. Set1 = {122, 2, 234, 54,……..}

✅ use of Dictionary

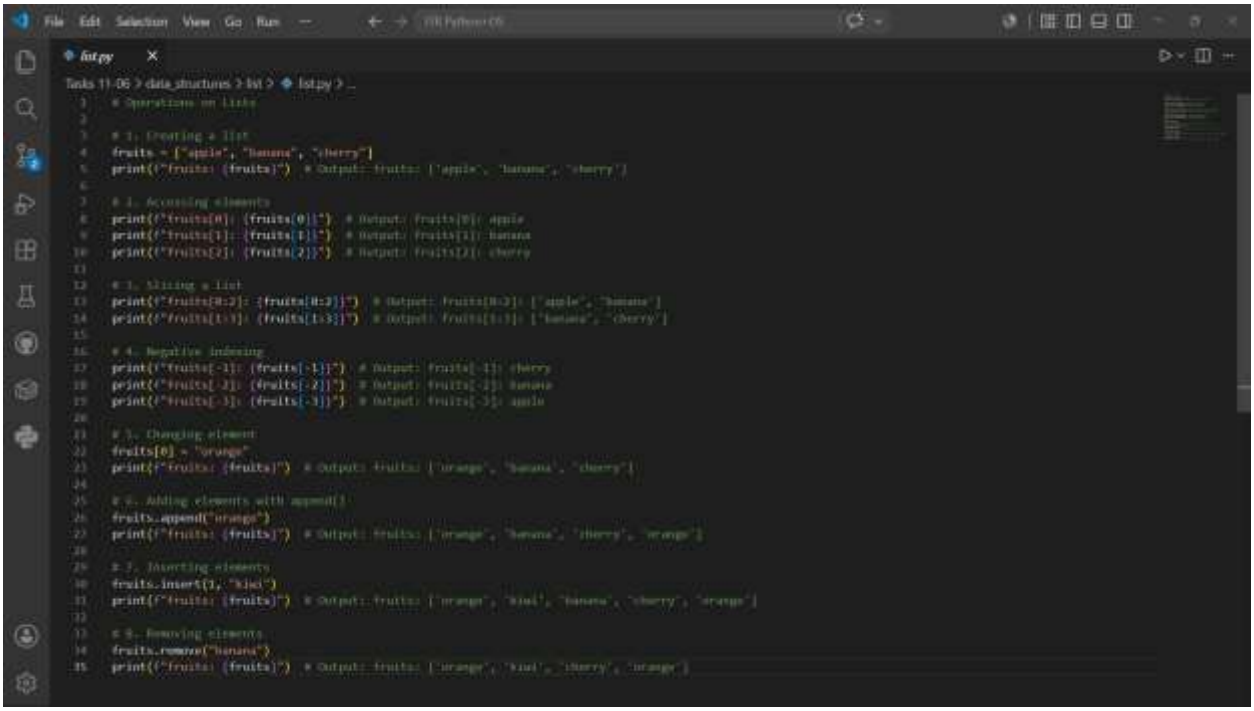
- Dictionaries are used when data is required in pairs of key-value. When heterogeneous data must be in relation with each other.

Eg. Dict1 = {“key1”:”value1”, ”key2”:”value2”,……..}

✅ Common method of all data structures

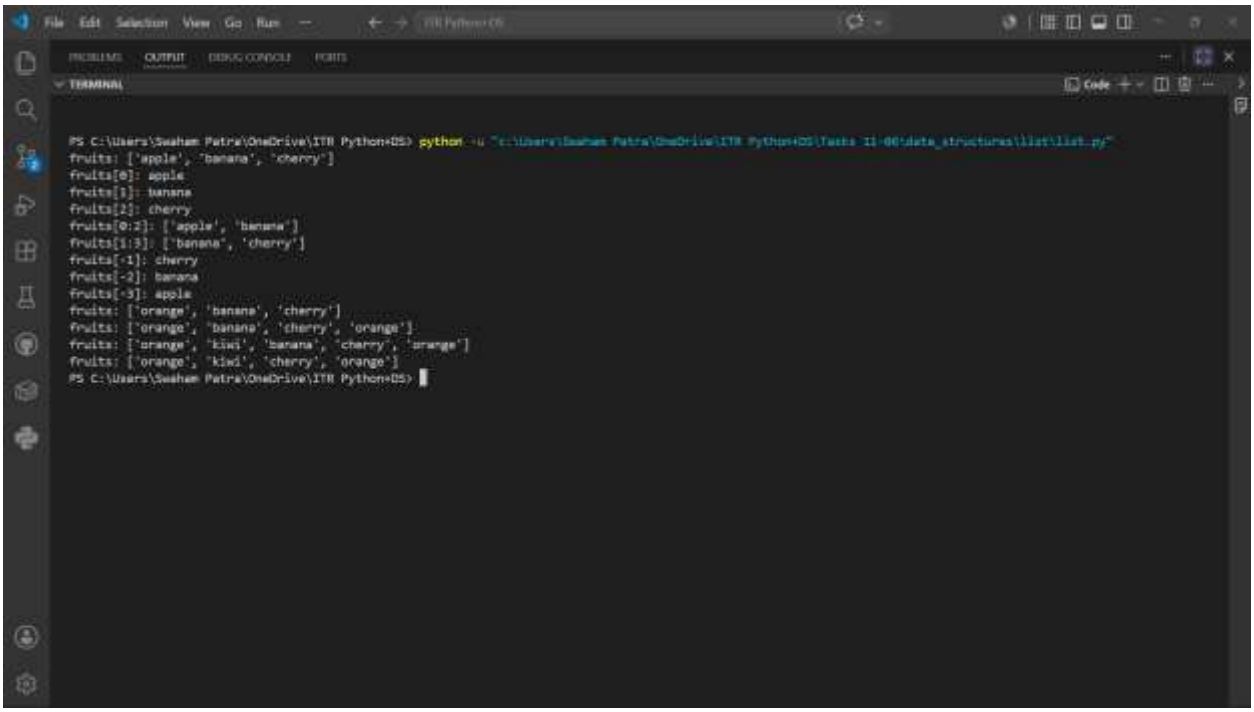
#List

##Code



```python
# Operations on Lists

# 1. Creating a list
fruits = ["apple", "banana", "cherry"]
print(f"Fruits: {fruits}")  # Output: Fruits: ['apple', 'banana', 'cherry']

# 2. Accessing elements
print(f"Fruits[0]: {fruits[0]}")  # Output: Fruits[0]: apple
print(f"Fruits[1]: {fruits[1]}")  # Output: Fruits[1]: banana
print(f"Fruits[2]: {fruits[2]}")  # Output: Fruits[2]: cherry

# 3. Slicing a list
print(f"Fruits[0:2]: {fruits[0:2]}")  # Output: Fruits[0:2]: ['apple', 'banana']
print(f"Fruits[1:3]: {fruits[1:3]}")  # Output: Fruits[1:3]: ['banana', 'cherry']

# 4. Negative indexing
print(f"Fruits[-1]: {fruits[-1]}")  # Output: Fruits[-1]: cherry
print(f"Fruits[-2]: {fruits[-2]}")  # Output: Fruits[-2]: banana
print(f"Fruits[-3]: {fruits[-3]}")  # Output: Fruits[-3]: apple

# 5. Changing element
fruits[0] = "orange"
print(f"Fruits: {fruits}")  # Output: Fruits: ['orange', 'banana', 'cherry']

# 6. Adding elements with append()
fruits.append("orange")
print(f"Fruits: {fruits}")  # Output: Fruits: ['orange', 'banana', 'cherry', 'orange']

# 7. Inserting elements
fruits.insert(1, "kiwi")
print(f"Fruits: {fruits}")  # Output: Fruits: ['orange', 'kiwi', 'banana', 'cherry', 'orange']

# 8. Removing elements
fruits.remove("banana")
print(f"Fruits: {fruits}")  # Output: Fruits: ['orange', 'kiwi', 'cherry', 'orange']
```

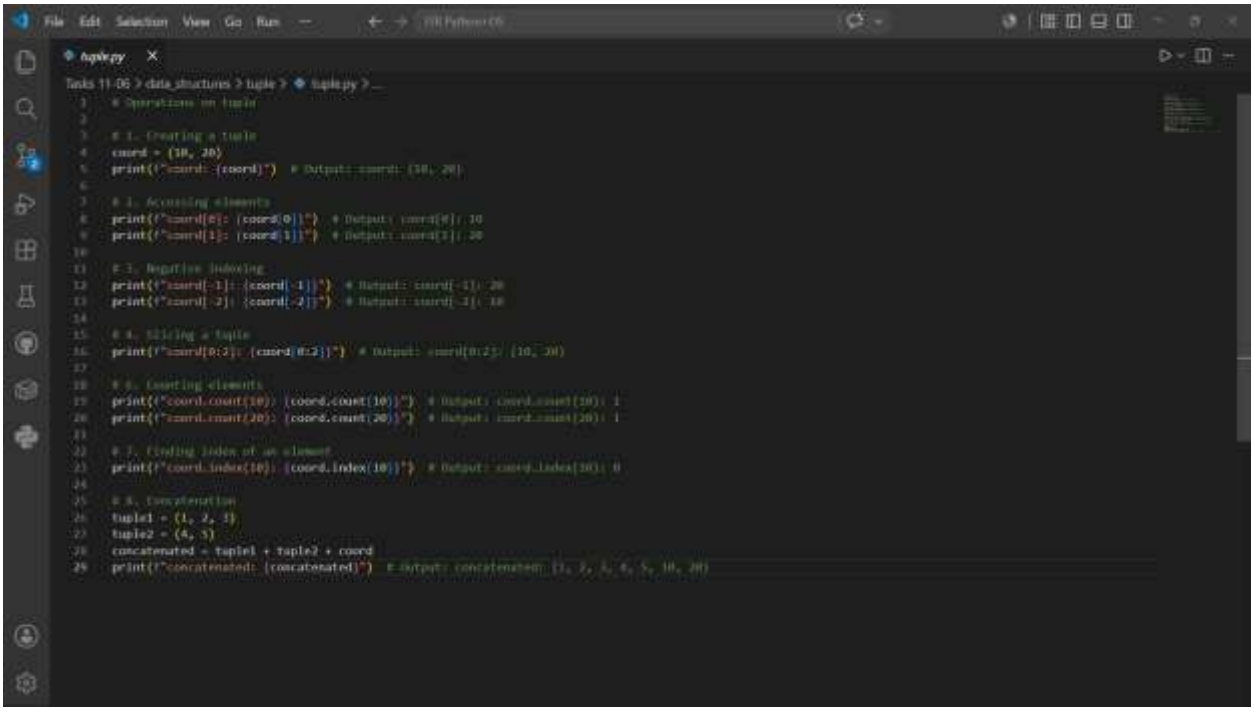
##Output



```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 11-06\data_structures\list\list.py"
Fruits: ['apple', 'banana', 'cherry']
Fruits[0]: apple
Fruits[1]: banana
Fruits[2]: cherry
Fruits[0:2]: ['apple', 'banana']
Fruits[1:3]: ['banana', 'cherry']
Fruits[-1]: cherry
Fruits[-2]: banana
Fruits[-3]: apple
Fruits: ['orange', 'banana', 'cherry']
Fruits: ['orange', 'banana', 'cherry', 'orange']
Fruits: ['orange', 'kiwi', 'banana', 'cherry', 'orange']
Fruits: ['orange', 'kiwi', 'cherry', 'orange']
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

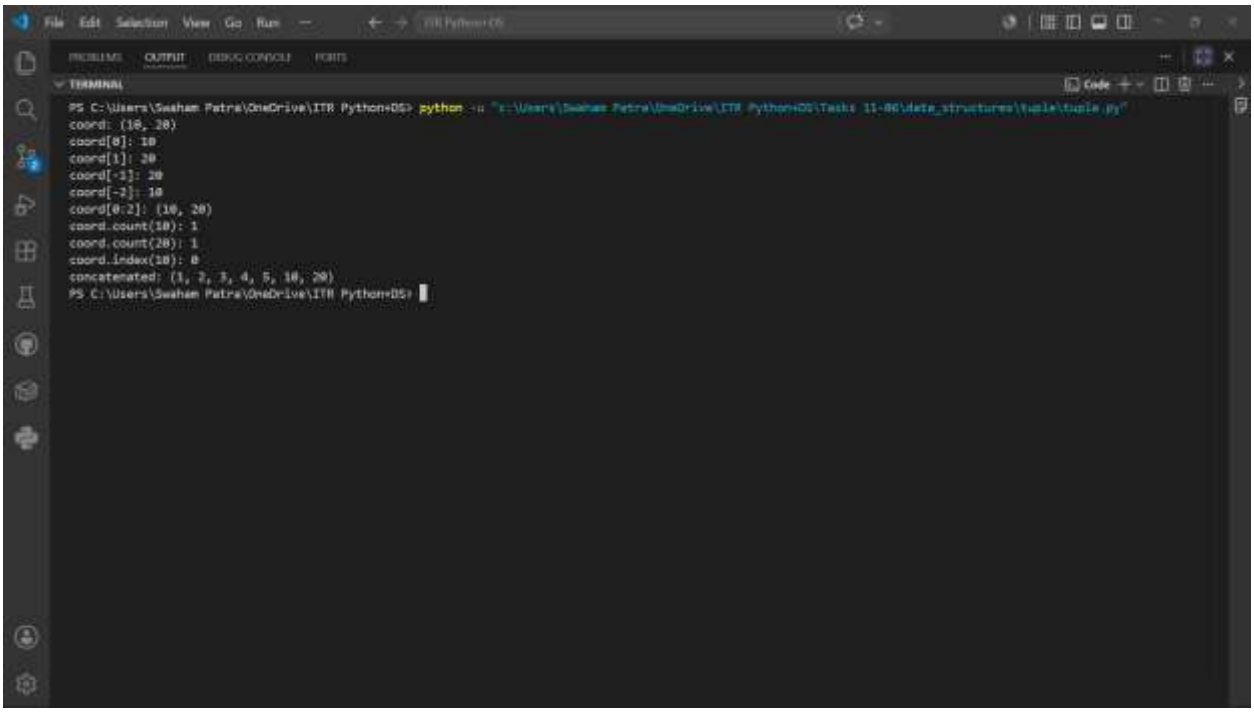
#Tuple

##Code



```
# Operations on tuple

# 1. Creating a tuple
coord = (10, 20)
print(f"coord: {coord}")  # Output: coord: (10, 20)

# 2. Accessing elements
print(f"coord[0]: {coord[0]}")  # Output: coord[0]: 10
print(f"coord[1]: {coord[1]}")  # Output: coord[1]: 20

# 3. Negative indexing
print(f"coord[-1]: {coord[-1]}")  # Output: coord[-1]: 20
print(f"coord[-2]: {coord[-2]}")  # Output: coord[-2]: 10

# 4. Slicing a tuple
print(f"coord[0:2]: {coord[0:2]}")  # Output: coord[0:2]: (10, 20)

# 5. Counting elements
print(f"coord.count(10): {coord.count(10)}")  # Output: coord.count(10): 1
print(f"coord.count(20): {coord.count(20)}")  # Output: coord.count(20): 1

# 7. Finding index of an element
print(f"coord.index(10): {coord.index(10)}")  # Output: coord.index(10): 0

# 8. Concatenation
tuple1 = (1, 2, 3)
tuple2 = (4, 5)
concatenated = tuple1 + tuple2 + coord
print(f"concatenated: {concatenated}")  # Output: concatenated: (1, 2, 3, 4, 5, 10, 20)
```

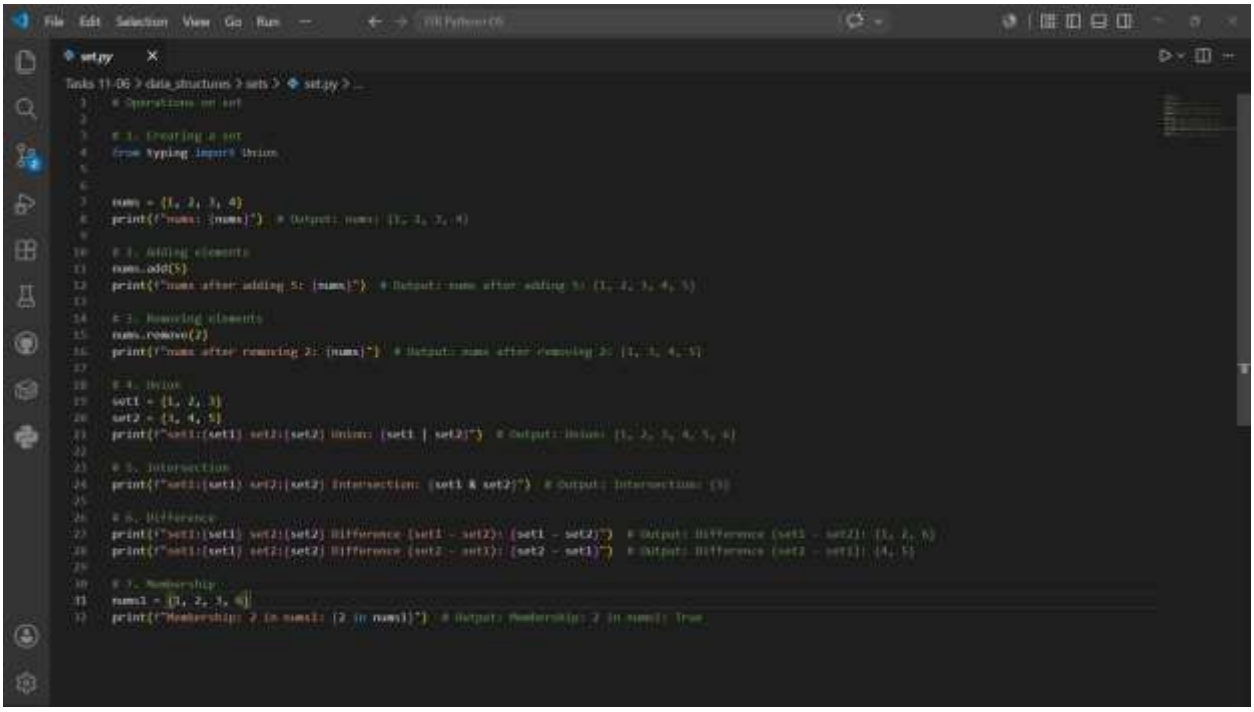
##Output



```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 11-06\data_structures\tuple\tuple.py"
coord: (10, 20)
coord[0]: 10
coord[1]: 20
coord[-1]: 20
coord[-2]: 10
coord[0:2]: (10, 20)
coord.count(10): 1
coord.count(20): 1
coord.index(10): 0
concatenated: (1, 2, 3, 4, 5, 10, 20)
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

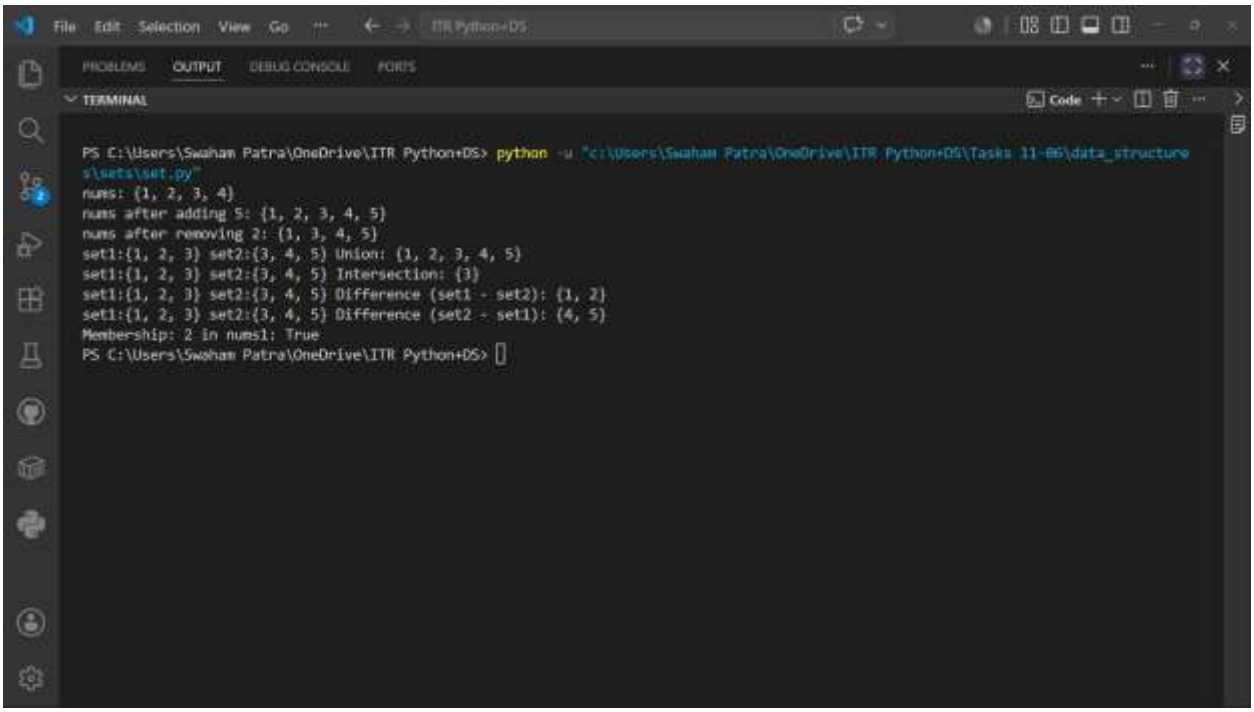
#Sets

##Code



```python
# Operations on set

# 1. Creating a set
from typing import Union


nums = {1, 2, 3, 4}
print(f"nums: {nums}")  # Output: nums: {1, 2, 3, 4}

# 2. Adding elements
nums.add(5)
print(f"nums after adding 5: {nums}")  # Output: nums after adding 5: {1, 2, 3, 4, 5}

# 3. Removing elements
nums.remove(2)
print(f"nums after removing 2: {nums}")  # Output: nums after removing 2: {1, 3, 4, 5}

# 4. Union
set1 = {1, 2, 3}
set2 = {3, 4, 5}
print(f"set1:{set1} set2:{set2} Union: {set1 | set2}")  # Output: Union: {1, 2, 3, 4, 5}

# 5. Intersection
print(f"set1:{set1} set2:{set2} Intersection: {set1 & set2}")  # Output: Intersection: {3}

# 6. Difference
print(f"set1:{set1} set2:{set2} Difference (set1 - set2): {set1 - set2}")  # Output: Difference (set1 - set2): {1, 2}
print(f"set1:{set1} set2:{set2} Difference (set2 - set1): {set2 - set1}")  # Output: Difference (set2 - set1): {4, 5}

# 7. Membership
nums1 = {1, 2, 3, 4}
print(f"Membership: 2 in nums1: {2 in nums1}")  # Output: Membership: 2 in nums1: True
```

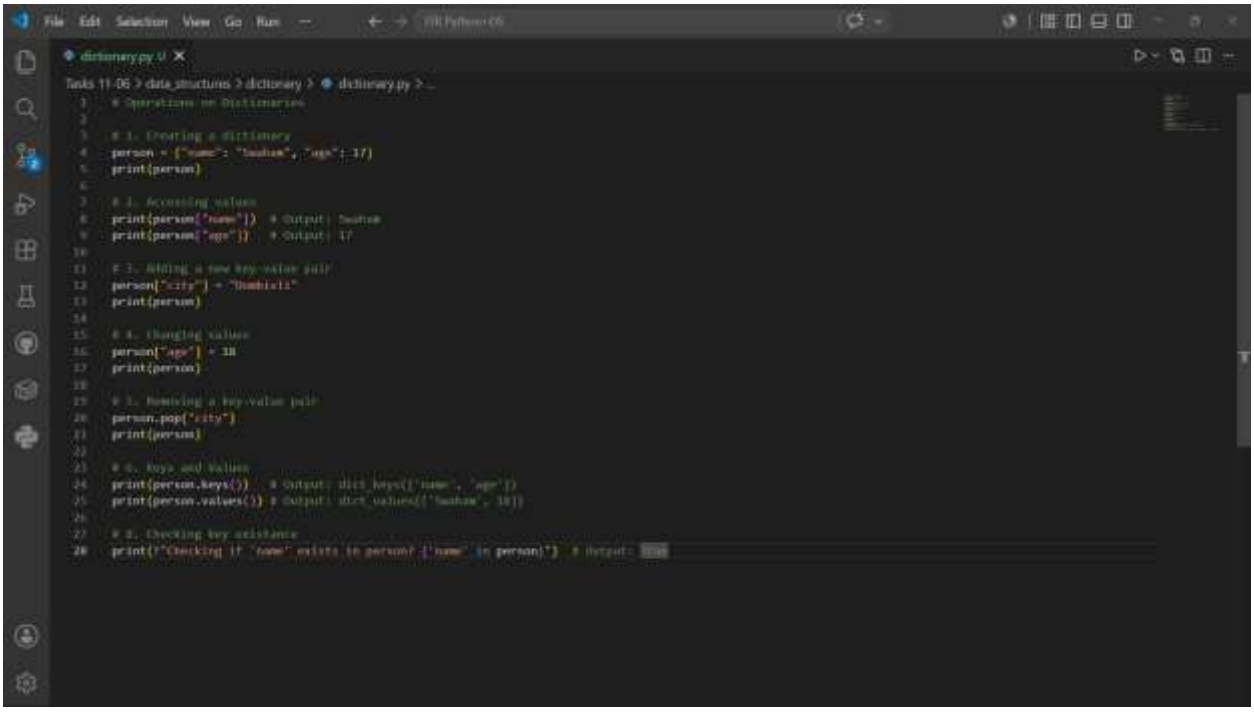
##Output



```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 11-06\data_structure
s\sets\set.py"
nums: {1, 2, 3, 4}
nums after adding 5: {1, 2, 3, 4, 5}
nums after removing 2: {1, 3, 4, 5}
set1:{1, 2, 3} set2:{3, 4, 5} Union: {1, 2, 3, 4, 5}
set1:{1, 2, 3} set2:{3, 4, 5} Intersection: {3}
set1:{1, 2, 3} set2:{3, 4, 5} Difference (set1 - set2): {1, 2}
set1:{1, 2, 3} set2:{3, 4, 5} Difference (set2 - set1): {4, 5}
Membership: 2 in nums1: True
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

#Dictionaries

##Code



##Output